SENSITIVITY ANALYSIS

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FROM PARTIAL IDENTIFICATION TO SENSITIVITY

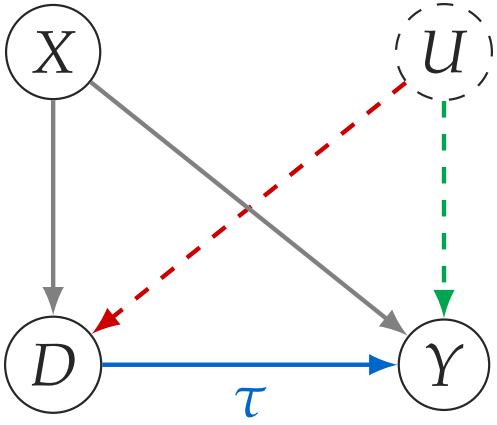
Partial Identification

- *What are the bounds under zero assumptions?*
- Sharp and Logical bounds
- In observational political science, the bounds can be uninformative: $0 \in [\tau_L, \tau_U]$

The Shift to Sensitivity Analysis

- Instead: *How severe should a violation of our identifying assumption be to make the estimate unreliable?*
- The main focus has been the impact of *Unobserved Confounders*.

THE PROBLEM: UNCONFOUNDEDNESS



- We want to estimate the Direct Path ($D \to Y$).
- We can block the path through X by conditioning on it.
- But U is *unobserved* (dashed). The Backdoor Path ($D \leftarrow U \to Y$) remains open.
- How strong should the red arrows be relative to the blue arrow to explain away our observed correlation?

THE PROBLEM: UNCONFOUNDEDNESS

- *Conditional Ignorability:*

$$Y(d) \perp\!\!\!\perp D \mid X \quad \text{for all } d \in \mathcal{D}$$

- *The Violation:*

$$Y(d) \not\perp\!\!\!\perp D \mid X \quad \text{but} \quad Y(d) \perp\!\!\!\perp D \mid X, U$$

- $\implies$ Quantify the changes in τ (The blue arrow) as a function of U, D, and Y (the red and green arrows)

OMITTED VARIABLE BIAS

- If U_i were observed, the true ATE (τ):

$$\tau = \mathbb{E}_U \left[\mathbb{E} \left[Y_i | D_i = 1, U_i \right] - \mathbb{E} \left[Y_i | D_i = 0, U_i \right] \right]$$

- But because U_i is unobserved, the naive estimator:

$$\hat{\tau} = \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]$$

- Then the OVB looks like:

$$Bias = \mathbb{E} \left[\hat{\tau} \right] - \tau$$

OMITTED VARIABLE BIAS

- By the Law of Iterated Expectations, the naive estimator $\hat{\tau}$:

$$
\begin{aligned}
\mathbb{E}[\hat{\tau}] &= \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] \\
&= \mathbb{E}_{U|D=1}\big[\mathbb{E}[Y_i \mid D_i = 1, U_i]\big] \\
&\quad - \mathbb{E}_{U|D=0}\big[\mathbb{E}[Y_i \mid D_i = 0, U_i]\big]
\end{aligned}
$$

$$
\begin{aligned}
Bias &= \mathbb{E}\left[\hat{\tau}\right] - \tau \\
&= \Big(\mathbb{E}_{U|D=1}\big[\mathbb{E}[Y_i \mid D_i = 1, U_i]\big] - \mathbb{E}_U\big[\mathbb{E}[Y_i \mid D_i = 1, U_i]\big]\Big) \\
&\quad - \Big(\mathbb{E}_{U|D=0}\big[\mathbb{E}[Y_i \mid D_i = 0, U_i]\big] - \mathbb{E}_U\big[\mathbb{E}[Y_i \mid D_i = 0, U_i]\big]\Big)
\end{aligned}
$$

OMITTED VARIABLE BIAS

- *Assumption (Constant Treatment Effect):*
  Assume the effect of D on Y is additive and does not depend on U:

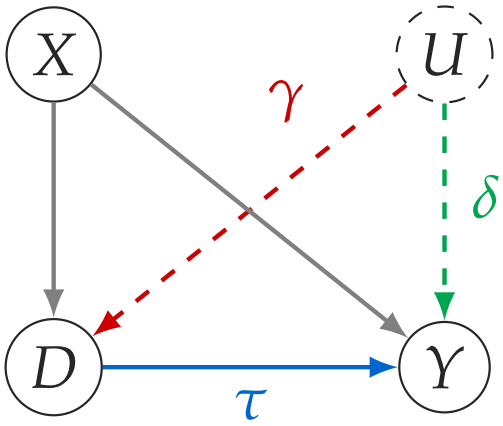
$$\mathbb{E}[Y_i \mid D_i, U_i] = \tau D_i + \gamma U_i$$

- Then,

$$\begin{aligned}
&\Big(\mathbb{E}_{U|D=1}\big[\mathbb{E}[Y_i \mid D_i = 1, U_i]\big] - \mathbb{E}_U\big[\mathbb{E}[Y_i \mid D_i = 1, U_i]\big]\Big) \\
&= \mathbb{E}_{U|D=1}[\tau + \gamma U_i] - \mathbb{E}_U[\tau + \gamma U_i] \\
&= \gamma \cdot \mathbb{E}[U_i \mid D_i = 1] - \gamma \cdot \mathbb{E}[U_i]
\end{aligned}$$

$$\begin{aligned}
Bias &= \gamma \cdot \{\mathbb{E}[U_i \mid D_i = 1] - \mathbb{E}[U_i \mid D_i = 0]\} \\
&= \gamma \cdot \delta
\end{aligned}$$

BACK THE DAG



- *Rosenbaum:* Focuses primarily on the *Selection* ($U \to D$).
- *Imbens:* Models both γ and δ using a hypothetical binary U.
- *Cinelli & Hazlett:* Models both γ and δ using variance explained (R^2).

APPROACH 1: ROSENBAUM (2002)

- *Focus:* The $U \to D$ relationship (Selection Bias).
- *The Logic:*
  - Imagine two people, i and j, who have the exact same observed X (e.g., same age, education, income).
  - In a perfect experiment, they have the same probability of being treated.
  - If there is a *Selection Bias*, one person might be more likely to get treatment because of their unobserved U.
- $\Longrightarrow$ How much more likely must i be to receive treatment than j (due to U) before our statistical significance ($p < 0.05$) disappears?

APPROACH 1: ROSENBAUM (2002)

- *Assumptions:* $D \in \{0, 1\}$ (binary treatment). Y can be any type.
- Let $\pi_i = P(D_i = 1 \mid X_i, U_i)$ be the true propensity score.
- For two units i and j with $X_i = X_j$, the odds ratio of being treated is bounded by Γ:

$$\frac{1}{\Gamma} \leq \frac{\pi_i/(1-\pi_i)}{\pi_j/(1-\pi_j)} \leq \Gamma$$

- *Parameters:*
  - $\Gamma = 1$: No hidden bias. U does not affect treatment assignment.
  - $\Gamma = 2$: One unit is twice as likely to be treated as the other due to U.
- *Testing Method:*
  1. Set a value for Γ.
  2. *Calculate the p-value bound:* Under H_0 (no effect), the test statistic T follows a distribution. With $\Gamma > 1$, we find the

APPROACH 1: ROSENBAUM (2002)

- *Consider S matched pairs. In each pair, one is treated ($D = 1$) and one is control ($D = 0$).*
- *1. The Probability of Treatment:*

  In any pair s, the probability that unit i being treated is:

$$p_s = \frac{\pi_i}{\pi_i + \pi_j}$$

- If $\Gamma = 1$, then $p_s = 0.5$. If $\Gamma > 1$, then $\frac{1}{1+\Gamma} \leq p_s \leq \frac{\Gamma}{1+\Gamma}$.
- *2. The Worst-Case Null Distribution:*
- To test $H_0 : \tau = 0$, we assume the "worst" possible p_s for our result.
  - If $\tau > 0$, then the "worst case" is that U increases $Y \rightarrow$ we use $p_s = \frac{\Gamma}{1+\Gamma}$
  - Calculate the p-value using the distribution of the sum of S independent Bernoulli variables with success probability p_s

APPROACH 2: IMBENS (2003)

- *Focus:* Both $U \to D$ and $U \to Y$ arrows simultaneously.
- *The Logic:*
  - Imbens asks: "What would happen if there was one specific binary variable U (like 'Low / High Motivation') that I forgot to include?"
  - We don't know how strongly U affects D or Y, so we test *all possible combinations* of those strengths
- $\implies$ What combination of γ and δ would make the point estimate of the τ exactly zero?

APPROACH 2: IMBENS (2003)

- *Assumptions:* $D \in \{0,1\}$, $U \in \{0,1\}$ is a *scalar, binary* confounder, and Y is continuous/binary
- Imbens assumes a parametric structure:
  1. *Selection* ($U \to D$): $P(D = 1 \mid X, U) = \Lambda(\alpha_0 + \alpha_X X + \gamma U)$
  2. *Outcome* ($U \to Y$): $Y = \tau D + \beta X + \delta U + \epsilon$
- *Method:*
  - We vary γ *(effect of U on treatment assignment)* and δ *(effect of U on outcome)*.
  - We plot a *contour map* showing the value of $\hat{\tau}$ for every pair of (γ, δ).

APPROACH 3: CINELLI & HAZLETT (2020)

- *Focus:* The *proportion of variance* explained by the Red Arrows.
- *The Logic:*
  - Instead of imagining a binary U, we think about *explanatory power*.
  - How much of the "mystery" in D and Y is explained by U?
- $\implies$ If U explains some of *the variation in treatment* and some of *the variation in the outcome*, is that enough to make $\hat{\tau}$ uninformative?
- R and STATA: `sensemakr`

APPROACH 3: CINELLI & HAZLETT (2020)

- *Assumptions:* Linear framework. D, Y, U can be *continuous, vectors, or any distribution*. U is assumed to be orthogonal to ϵ.
- C&H parameterize the bias using *Partial* R^2 (unit-free variance explained):
  - $R^2_{D \sim U|X}$: The proportion of the variance in treatment D explained by U (after accounting for X).
  - $R^2_{Y \sim U|D,X}$: The proportion of the variance in outcome Y explained by U (after accounting for D and X).
- *The Exact Bias Formula:*

$$|\hat{\tau}_{res} - \tau| = \text{se}(\hat{\tau}_{res})\sqrt{\text{df}}\sqrt{\frac{R^2_{Y \sim U|D,X} R^2_{D \sim U|X}}{1 - R^2_{D \sim U|X}}}$$

CINELLI & HAZLETT: THE ROBUSTNESS VALUE

- *Robustness Value (RV)* simplifies the problem to: *"If an unobserved variable U was **equally related to the treatment and the outcome**, how strong would it have to be to nullify my result?"*
- *The RV is a "Safety Margin":*
  - If your RV is 0.20, your result can "carry the weight" of any confounder that explains up to 20% of the variance in D and Y
  - If a confounder exceeds this 20% "budget," then it exceeds the budget and results becomes insignificant
- *Summarize:* High RV $\implies$ a confounder must be very powerful in *both* dimensions to flip your conclusion

CINELLI & HAZLETT: CALCULATING THE RV

- Equal strength in both dimensions: $R^2_{Y \sim U|D,X}$ = $R^2_{D \sim U|X}$ = R^2
- Let $f_q = q \cdot \frac{|\hat{\tau}_{res}|}{\text{se}(\hat{\tau}_{res})\sqrt{\text{df}}}$, where $q \in (0, 1]$ is the *target reduction* of the estimate (e.g., $q = 1$ to reduce $\hat{\tau}$ to zero).
- The bias formula simplifies to a function of R^2:

$$f_q = \frac{R^2}{\sqrt{1 - R^2}}$$

- Solving for R^2 yields the *Robustness Value*:

$$RV_q = \frac{1}{2}\left(\sqrt{f_q^4 + 4f_q^2} - f_q^2\right)$$

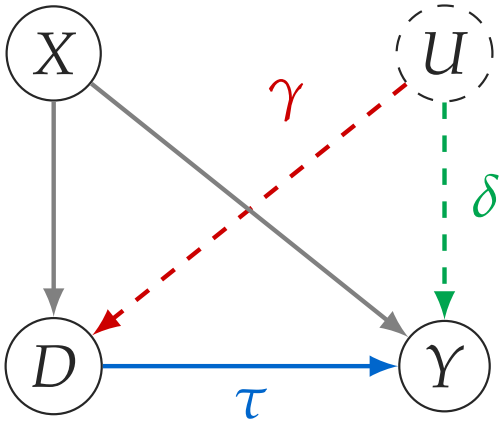
FORMAL BENCHMARKING (ALTONJI, ELDER, TABER LOGIC)

- *How large should an RV be to be "big"?*
  $\implies$ Compare U to an observed covariate X_j
- We assume U is k times as "strong" as X_j in its relationship to treatment and outcome:

$$R^2_{Y \sim U | D, X_{-j}} \leq k \cdot R^2_{Y \sim X_j | D, X_{-j}}$$
$$R^2_{D \sim U | X_{-j}} \leq k \cdot R^2_{D \sim X_j | X_{-j}}$$

- *The "k-multiplier" Argument:*
  - If $k = 1$, U is as powerful as X_j.
  - *Example:* "My RV is 0.15, but my strongest control (GDP) only has an R^2 of 0.05. Thus, an unobserved confounder would have to be *3x as strong as GDP* to nullify my result."

SUMMARY: SENSITIVITY TO UNOBSERVED CONFOUNDING



- Sensitivity analysis *does not prove* unconfoundedness! it *quantifies the cost of its failure*
- To move the conversation from *"Is there bias?"* to *"Is the bias large enough to change my substantive conclusion?"*

BEYOND UNCONFOUNDEDNESS: OTHER ASSUMPTIONS

- *Instrument Validity (Exclusion Restriction):*
  - *Assumption:* $Z \perp\!\!\!\perp Y \mid D, X$. The instrument only affects Y through D.
  - *Sensitivity:* Allow a direct effect ρ of Z on Y (Conley et al., 2012)
- *Parallel Trends (Difference-in-Differences):*
  - *Assumption:* Counterfactual trends are identical.
  - *Sensitivity:* Allow for specific violations of the trend (e.g., linear or quadratic) (Rambachan & Roth, 2023).